The ProLith Project: Developing Lithium Extraction in Namibia

Collaborative EU Funding Initiative for the Uis Tin Mine

UIS PETALITE

LOWER-INTENSITY CONVERSION PATHWAY

BATTERY-GRADE LITHIUM MATERIALS

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UIS PETALITE • ANDRADA MINING × GTK

Project Overview and Strategic Partnership

Three complementary partners aligning resources, research, and technology around petalite conversion.

Andrada Mining

Uis lithium resource, Operating Mine, and implementation platform

GTK

EU funding application coordination, applied mineralogy, and research integration

Metso

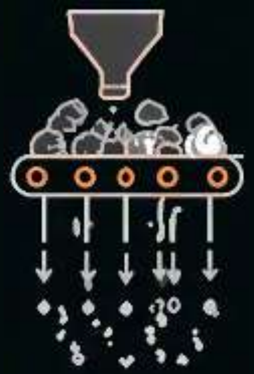
Sodium/alkaline leach expertise and industrial process technology

Uis petalite → sustainable conversion pathway → battery-material optionality

ProLith initiative:
Focused on sustainable, scalable petalite conversion from Namibia.

Grade and Recovery Evaluation

Value is optimised across the whole flowsheet - not by concentrate grade alone.



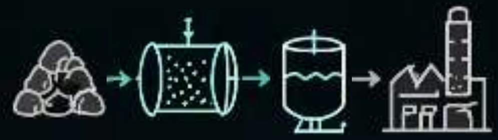
Ceramic-grade target

4.2% Li_2O

Higher concentrate specification
Fewer contaminants

Shift from grade only target to whole-system considerations

value



ProLith conversion target

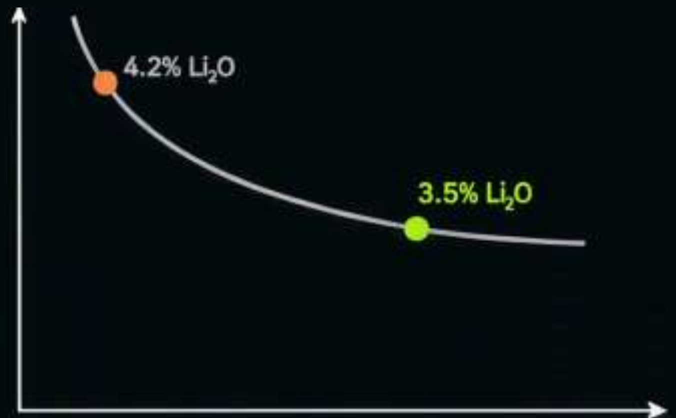
3.5% Li_2O

Optimise recovery + conversion response

Less stringent concentrate specifications

Reduced beneficiation intensity

Increased lithium- recovery
Reduced processing costs

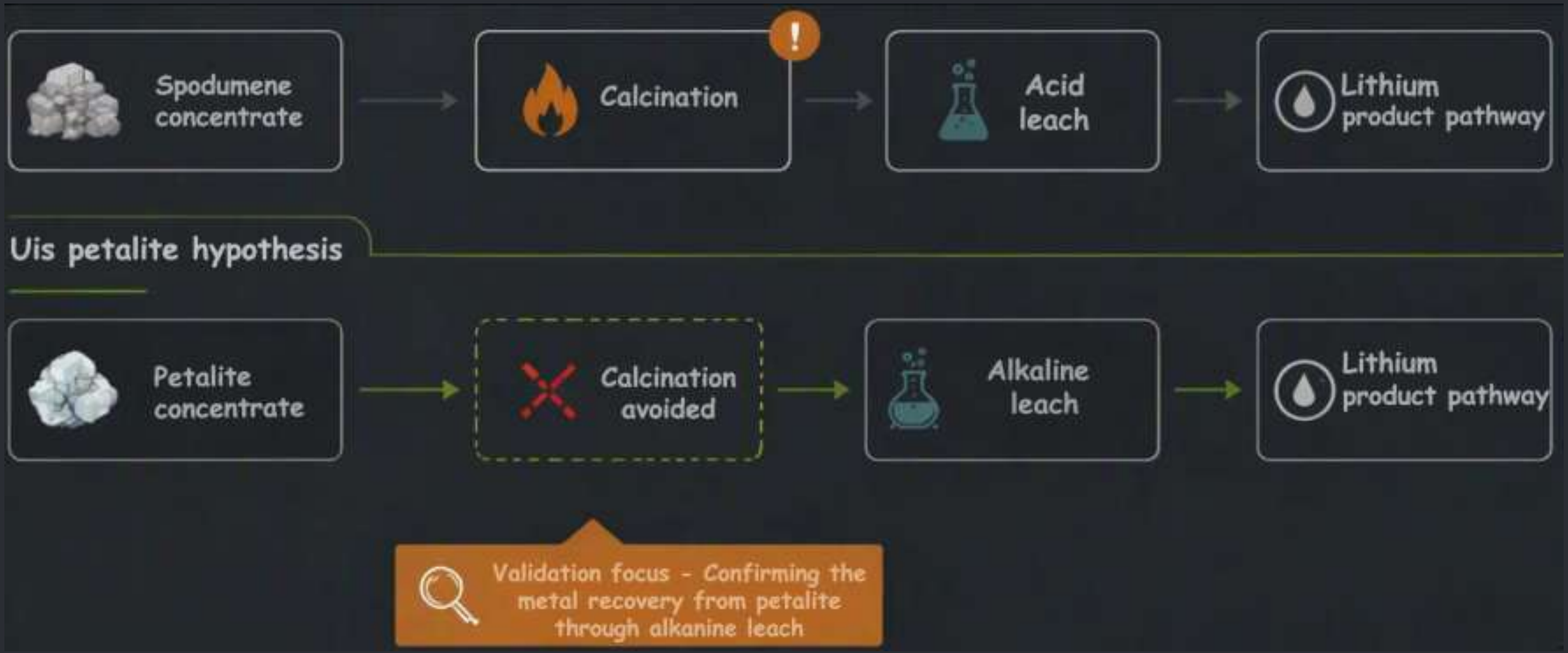


The Traditional Metallurgical Challenge

Validation focus: assessing whether Uis petalite can be processed directly without a calcination stage.

Spodumene route

Energy- Intensive
Thermal Activation



Potential enabler

Removing petalite calcination will lower process intensity and operating burden — test results to confirm.

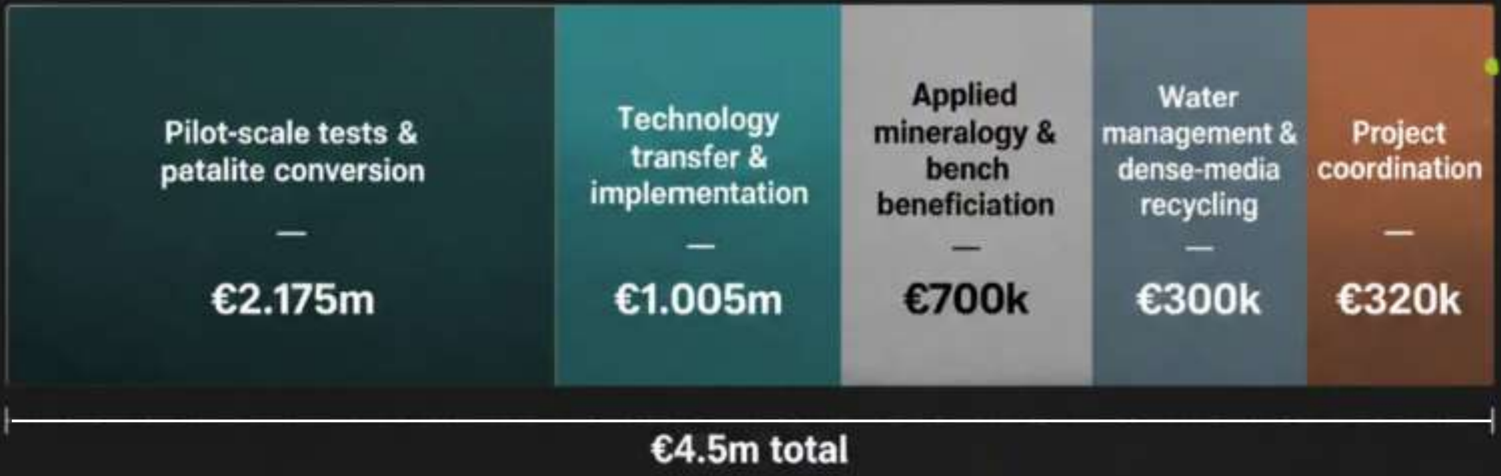
- Sodium / alkaline leach targets selective lithium extraction
- Petalite direct processing is the Prolith hypothesis test
- Spodumene still requires conversion before Alkaline Leaching
- Recycle opportunities may support a lower-intensity flowsheet

Scope of the EU Funding Allocation

Objective: To Generate Mineralogical, Metallurgical, and Processing Data

EU funding application in **progress**

Indicative programme budget: €4.5m



- Pilot-scale tests & petalite conversion — €2.175m
- Technology transfer & implementation — €1.005m
- Applied mineralogy & bench beneficiation — €700k
- Water management & dense-media recycling — €300k
- Project coordination — €320k

Funding is targeted toward evidence generation, scale-up readiness, and site applicability — not a claim of completed commercial performance.

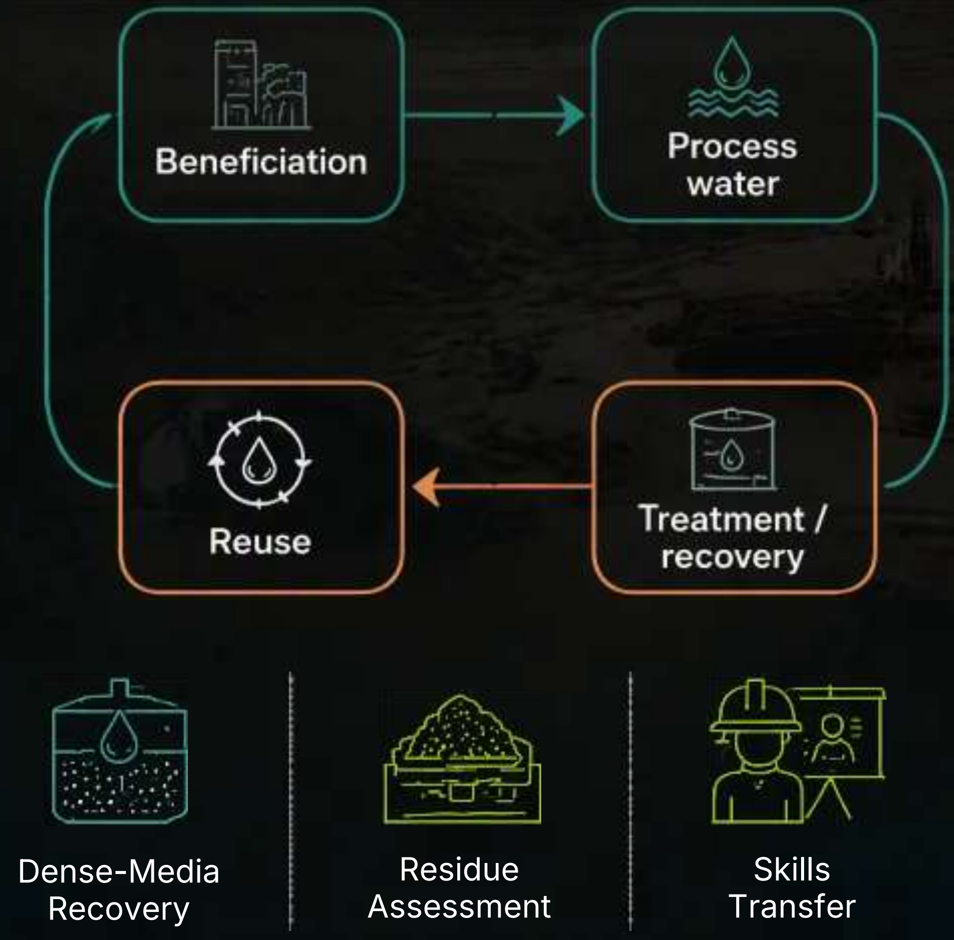
Feed-grade test matrix

Relates initial feed grade to recovery, conversion response, water recovery and optimisation

- Feed grade
- Recovery
- Impurity deportment
- Product pathway

Environmental and Social Governance Focus

- Closed-loop process-water optimisation for an arid operating environment
- Dense-media rinsing, recovery and recycling investigation
- Residue and by-products assessed for recovery, reuse or safe disposal
- Vocational training and skills transfer for Namibian operating teams



Maximizing value while minimizing impact

Future Vision: Phase 2

From petalite validation to circular resource management at Uis

Future ambition — Eliminating Waste

Phase 1: Validate

Petalite recovery, conversion response, water, media optimisation and skills transfer

Phase 2: Integrate

Circular resource management across Uis material streams

Long-Term Aspiration

Recovery, reuse or responsible disposition toward a zero-waste Uis mine

Phase 1 investigation:
feed-grade response, recovery performance, residue behaviour and site applicability.

Potential Impact of the ProLith Project

Shared value for Namibia and the European Union

Potential outcomes - subject to successful technical validation and commercial implementation

NAMIBIA

Battery-market access - Uis could produce petalite concentrate suitable for the battery-material value chain.

Increased company revenues - a new petalite product stream could capture more value from mined material.

Additional tax revenue - stronger operating revenues could increase fiscal contributions to Namibia.

Additional employment - petalite production and processing will generate additional jobs.

Proof of concept

➔

industrial deployment

Finnish technology deployed - successful validation could lead to the deployments of Finnish processing technology and expertise.

Full-scale facility opportunity - positive result would support procurement of a commercial petalite-conversion facility.

EU pegmatite value unlocked - proof of concept could open a development pathway for EU pegmatite projects containing petalite.

Conclusion & Q&A

Lower Process Intensity

Validation focus: Utilisation of petalite for the production of lithium carbonate

Higher Resource Recovery

Optimize around a 3.5% Li_2O conversion feed.

Namibian Value Addition

Build local capability through skills transfer, site advancement, and technology integration.

Can Uis set a new benchmark for sustainable petalite conversion?

Questions & discussion

Andrada Mining × GTK